

1 Semester Test 1

L 2024

①

Memo

(Q1)

$$\begin{aligned} Q_e &= I \times t \\ &= 10 \times 10^3 \times 15 \times 10^{-6} \\ &= 0,15 \text{ C or } 150 \text{ mC} \end{aligned}$$

(Q2)

$$\begin{aligned} W_e &= Q_e \times V \\ &= 0,15 \times 1 \times 10^9 \\ &= 1,5 \times 10^8 \text{ J} \\ &= (1,5 \times 10^8 / 3600) \text{ Wh} \\ &= 41667,67 \text{ Wh or } 41,6667 \text{ kWh} \end{aligned}$$

(2)

$$(Q3) \quad \sum \text{Power in circuit} = 0$$

$$-20W + P_s + 10W + 20W + 30W = 0$$

$$P_s = -40W$$

(Q4)

$$P_{R_1} = 10W$$

$$= (I_{R_1})^2 R_1$$

$$= (0,2)^2 R_1$$

$$R_1 = 250 \Omega$$

Q5) KCL at node above 3Ω resistor :

③

$$i_1 + i_3 - 3i_1 = 0$$

$$i_3 = 2i_1 \quad \text{--- ①}$$

KVL left loop :

$$7 - i_1 + 3i_3 = 0$$

$$7 - i_1 + 6i_1 = 0$$

$$i_1 = -1,4 \text{ A}$$

KVL right loop :

$$3i_3 - V_1 + 5 \times 3i_1 = 0$$

$$6i_1 - V_1 + 15i_1 = 0$$

$$V_1 = 21i_1$$

$$= -29,4 \text{ V}$$

$$\text{Q6) } P = (i_1)^2 \cdot R_1$$

$$= 1,96 \text{ W}$$

Q7) Voltage divider :

$$\begin{aligned}
 V_1 &= V_s \left(\frac{3}{3 + 1 \parallel 2} \right) \\
 &= V_s \left(\frac{3}{3 + \frac{2}{3}} \right) \\
 &= 11 \left(\frac{9}{11} \right) \\
 &= 9 \text{ V}
 \end{aligned}$$

Q8) $V_{1\Omega} = 1 \times I_1 = 1 \text{ V}$

$$V_{2\Omega} = V_{1\Omega} = 1 \text{ V}$$

$$I_2 = \frac{V_{2\Omega}}{2\Omega} = 0,5 \text{ A}$$

$$I_3 = I_1 + I_2 \quad (\text{KCL})$$

$$I_3 = 1,5 \text{ A}$$

KVL bottom loop :

$$V_s - V_{3\Omega} - V_{2\Omega} = 0$$

$$V_s - 1,5 \cdot 3 - 1 = 0$$

$$V_s = 5,5 \text{ V}$$

$$Q9) I_0 = \frac{10V}{1k\Omega + 2k\Omega} = \frac{10}{3} \text{ mA}$$

(5)

$$V_1 = 2k\Omega \times I_0$$

$$= \frac{2 \cdot 10}{3}$$

$$= 6,667 \text{ V}$$

Q10) Current divider :

$$I_3 = 150 I_0 \left(\frac{4k\Omega}{3k\Omega + 4k\Omega} \right)$$

$$= 150 \left(\frac{10}{3} \right) \left(\frac{4}{7} \right) \text{ mA}$$

$$= 150 \left(\frac{10}{3000} \right) \left(\frac{4}{7} \right) \text{ A}$$

$$= \frac{1}{2} \cdot \frac{4}{7} \text{ A}$$

$$= \frac{2}{7} \text{ A}$$

$$= 0,2857 \text{ A}$$

(6)

$$\begin{aligned} Q_{11) \quad R_{eq} &= 10 + (6 + R \parallel R \parallel R) \parallel 10 \\ &= 10 + (6 + \frac{R}{3}) \parallel 10 \\ &= 15 \Omega \end{aligned}$$

$$\therefore 6 + \frac{R}{3} = 10$$

$$R = 12 \Omega$$

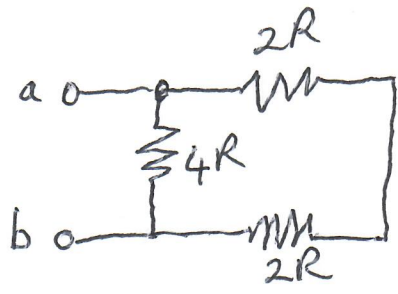
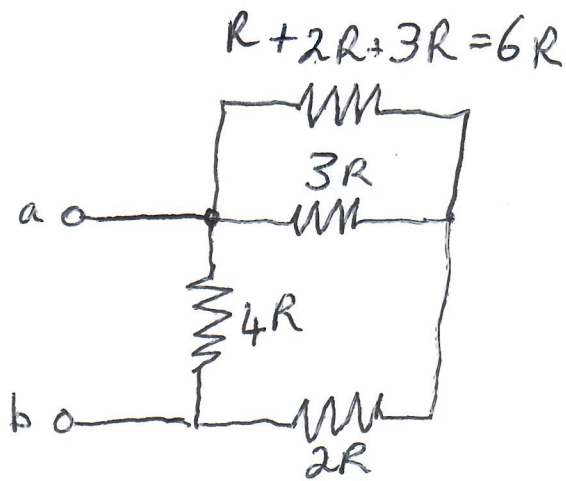
Q₁₂) Current divider :

$$I_a = I_a \left(\frac{\frac{1}{R}}{\frac{1}{R} + \frac{1}{R} + \frac{1}{R}} \right)$$

$$9 = I_a \left(\frac{1}{3} \right)$$

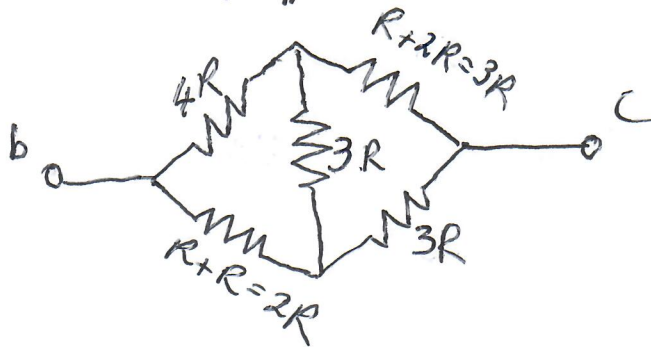
$$I_a = 27 \text{ A}$$

Q13)

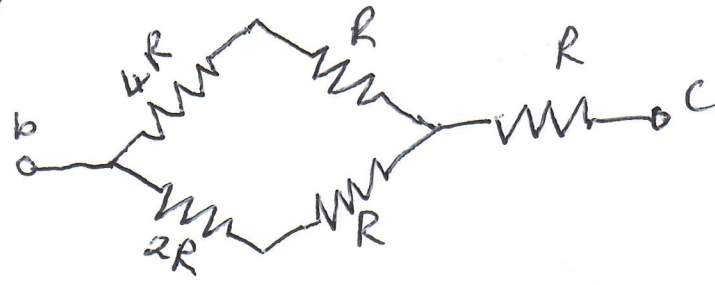


$$R_{ab} = 4R \parallel 4R = 2R = 2 \text{ k}\Omega$$

Q14)



Δ -to-Y :



$$\begin{aligned} R_{cb} &= (4R+R) \parallel (2R+R) + R \\ &= 2,875 R \\ &= 2,875 \text{ k}\Omega \end{aligned}$$

Q15 to Q18)

8

$$V_2 = -6 \text{ V} \quad \text{--- (1)}$$

KCL at V_3 node :

$$\frac{V_3 - V_2}{5} + \frac{V_3 - V_1}{8+7} + 4 = 0$$

$$3V_3 - 3V_2 + V_3 - V_1 + 60 = 0$$

$$3V_3 + 18 + V_3 - V_1 + 60 = 0$$

$$-V_1 + 4V_3 = -78 \quad \text{--- (2)}$$

KCL at V_1 node :

$$\frac{0 - V_1}{6} + \frac{V_2 + 12 - V_1}{10} + \frac{V_3 - V_1}{15} = 0$$

$$-\frac{V_1}{6} + \frac{6 - V_1}{10} + \frac{V_3 - V_1}{15} = 0$$

$$-15V_1 + 54 - 9V_1 + 6V_3 - 6V_1 = 0$$

$$-30V_1 + 6V_3 = -54 \quad \text{--- (3)}$$

Q15) $a = -30$

Q16) $b = -54$

Q17) $c = -1$

Q18) $d = 4$

Q19 to Q22)

(9)

$$V_3 + 12 = V_2 \quad \text{--- (1)}$$

$$V_1 = 3V_2 \quad \text{--- (2)}$$

$$\text{From (1): } V_2 - V_3 = 12 \quad \text{--- (3)}$$

KCL at V_2/V_3 node:

$$\frac{V_1 - V_2}{12k} + \frac{0 - V_2}{4k} + \frac{V_1 - V_3}{12k} + \frac{0 - V_3}{6k} \quad \text{--- (4)}$$

(2) in (4):

$$\frac{3V_2 - V_2}{12k} + \frac{0 - V_2}{4k} + \frac{3V_2 - V_3}{12k} + \frac{0 - V_3}{6k}$$

$$2V_2 - 3V_2 + 3V_2 - V_3 - 2V_3 = 0$$

$$2V_2 - 3V_3 = 0$$

$$-2V_2 + 3V_3 = 0 \quad \text{--- (5)}$$

$$\text{Q19) } j = -2$$

$$\text{Q20) } k = 0$$

$$\text{Q21) } l = 1$$

$$\text{Q22) } m = -1$$

Q23 to Q26)

(10)

$$V_x = 5i_2 \quad \text{--- (1)}$$

$$i_1 = 2V_x = 10i_2 \quad \text{--- (2)}$$

KVL bottom left loop :

$$3 + 5i_2 + 7(i_2 - i_1) + 3(i_2 - i_3) = 0$$

$$3 + 5i_2 + 7(i_2 - 10i_2) + 3(i_2 - i_3) = 0$$

$$3 + 5i_2 - 63i_2 + 3i_2 - 3i_3 = 0$$

$$3 - 55i_2 - 3i_3 = 0$$

$$55i_2 + 3i_3 = 3 \quad \text{--- (3)}$$

KVL bottom right loop :

$$3(i_3 - i_2) - 5 + 2i_3 = 0$$

$$3i_3 - 3i_2 + 2i_3 = 5$$

$$5i_3 - 3i_2 = 5$$

$$-3i_2 + 5i_3 = 5$$

$$Q23) e = -3$$

$$Q24) f = 5$$

$$Q25) g = 55$$

$$Q26) h = 3$$

Q27 to Q30)

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$$i_1 - i_2 = 2A \quad \text{--- (1)}$$

KVL around outer loop :

$$i_1 (50 + 25 + 125) - 15 + 50 i_2 = 0$$

$$200 i_1 + 50 i_2 = 15 \quad \text{--- (2)}$$

$$\text{Q27) } W = 200$$

$$\text{Q28) } x = 50$$

$$\text{Q29) } y = 1$$

$$\text{Q30) } z = -1$$

Q31 to Q34)

12

$$\begin{bmatrix} 16 & -8 \\ -8 & 14 \\ -5 & -4 \end{bmatrix} \begin{bmatrix} \hat{i}_1 \\ \hat{i}_2 \\ \hat{i}_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -15 \\ -30 \end{bmatrix}$$

Q31) $p = 16$

Q32) $q = -8$

Q33) $r = -4$

Q34) $s = -15$

(Q35)

13

$$\begin{aligned} I_E &= (1 + \beta) I_B \\ &= 100 \times 10^{-6} \times 100 \\ &= 10 \text{ mA} \\ &= 0,01 \text{ A} \end{aligned}$$

$$\begin{aligned} V_{RE} &= 400 \cdot I_E \\ &= 4 \text{ V} \end{aligned}$$

KVL input loop :

$$5,7 - I_B \cdot R - V_{BE} - V_{RE} = 0$$

$$5,7 - 100 \times 10^{-6} R - 0,7 - 4 = 0$$

$$\begin{aligned} R &= \frac{1}{100 \times 10^{-6}} = 10000 \Omega \\ &= 10 \text{ k}\Omega \end{aligned}$$

(Q36) $I_C = \beta \cdot I_B = 99 \times 100 \times 10^{-6} \text{ A}$

KCL at collector :

$$i_{R500} + i_{R600} + I_C = 0$$

$$\frac{V_0 - 0}{600} + \frac{V_0 - 0}{500} + I_C = 0$$

$$V_0 = 6,391 \text{ V}$$